Cognitive Market Research & Consulting — Industrial Practice

Commercial & Industrial RO Membrane Systems 2025

High-Recovery Separation, Zero Liquid Discharge (ZLD), and Industrial Process Water

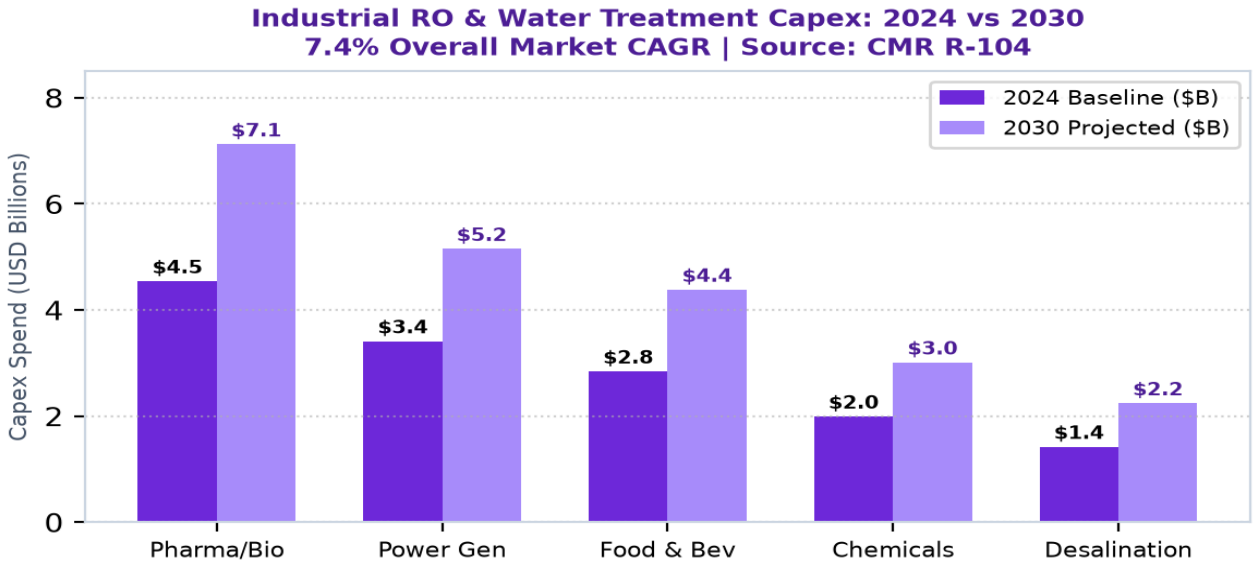
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Executive Overview

The global commercial and industrial reverse osmosis systems market was valued at USD 14.20 billion in 2024 and is projected to expand at a 7.4% CAGR through 2030. Strict industrial wastewater discharge mandates and clean energy hydrogen electrolysis feed requirements are catalyzing major capital expenditures across high-purity sectors.

1. Sector Capex Breakdown

Pharmaceutical manufacturing and power generation demand ultra-pure water with stringent conductivity limits, accounting for the largest share of industrial membrane capital spend.



North America represents 36% of global industrial expenditure, bolstered by semiconductor cleanroom expansions and petrochemical retrofits.

2. Technology Evolution: High-Pressure Brackish & Seawater RO

Next-generation polyamide thin-film composite (TFC) membranes feature anti-fouling coatings that extend operating lifespans by up to 35% in high-salinity industrial wastewater streams.